

Epithelial Tissue

Department of Anatomy

- Histology(2)

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Lecturer points:

- Most organs can be divided into?
(Parenchyma and Stroma).
- Structure of Epithelia.
- Basement Membranes.
 - Example: kidney glomeruli the basement membrane
- Intercellular Adhesion & Other Junctions.
- The principal functions of epithelial tissues.
- Epithelia can be divided into two main groups.
 - **Covering (or lining) epithelia.**
 - **Secretory (glandular) epithelia.**
- Covering (or lining) epithelia.
- Classification of covering or Lining Epithelia.
- Types of epithelia.
- Common types of covering epithelia

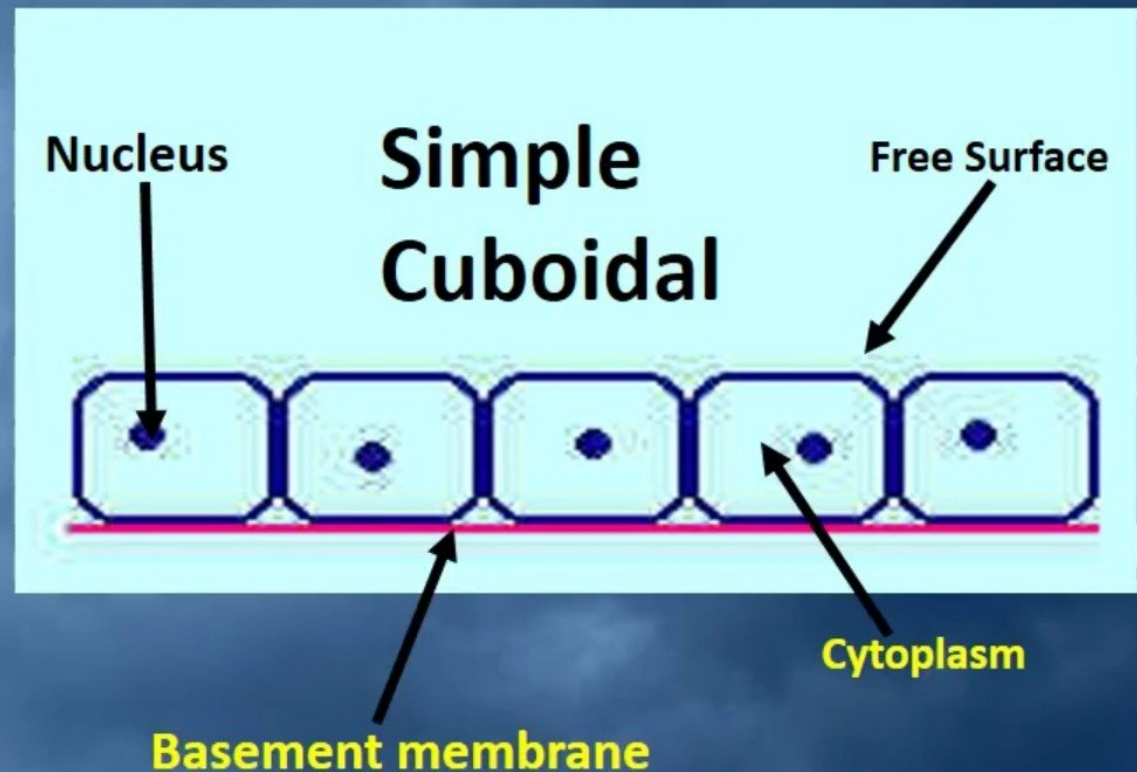
Epithelia can be divided into two main groups:

- 1) Covering (or lining) epithelia.**
- 2) Secretory (glandular) epithelia.**

Covering (or lining) epithelia

Covers surfaces and lines cavities epithelia

- Has a **free surface**.
- Other **surface attached** to underlying tissue by the **basement membrane**.



Classification of covering or Lining Epithelia

- Cells of **covering epithelia** are organized into **one** or **more layers** that **cover** the **external surface** or **line the cavities** of an organ.
- Epithelia are classified according to:
 - 1) The number of cell layers.
 - 2) The cell morphology in the surface layer.

Classification of covering or Lining Epithelia

- Epithelia are **classified according to the number of cell layers** and **(the cell morphology in the surface layer)**:
- **Simple epithelia** contain **one cell layer**.
- **Stratified epithelia** contain **two or more layers**.

Classification of covering or Lining Epithelia

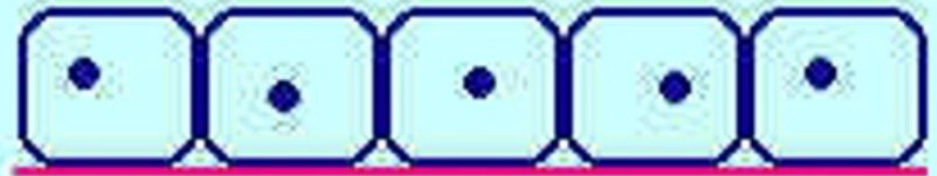
Based on cell shape simple epithelia are classified as:

- **Squamous** (thin cells).
- **Cuboidal** (cell width and thickness roughly similar).
- **Columnar** (cells taller than they are wide).

Simple
Squamous



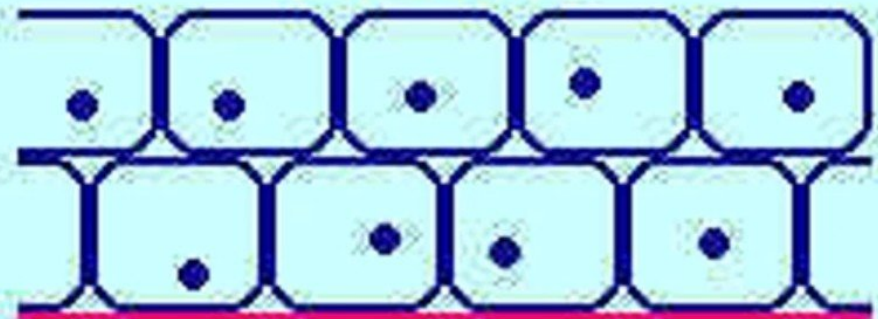
Simple
Cuboidal



Stratified
Squamous



Stratified
Cuboidal



Type of Epithelium

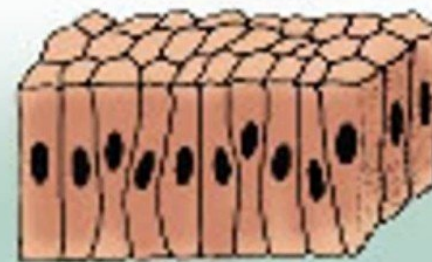
Types of Epithelium



Simple squamous



Simple cuboidal



Simple columnar

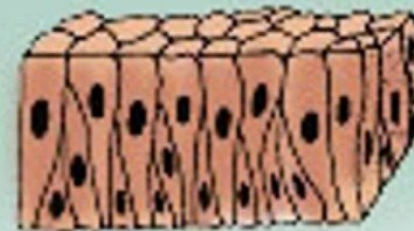
Transitional



Stratified squamous



Stratified cuboidal

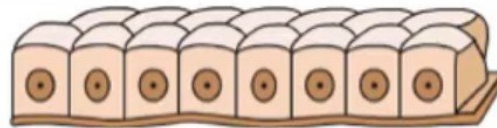


Pseudostratified columnar

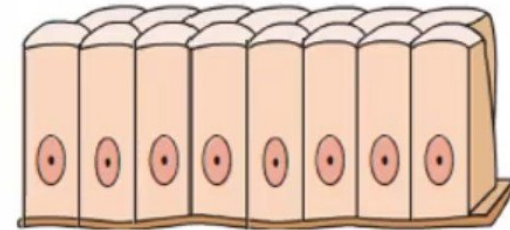
Type of Epithelium



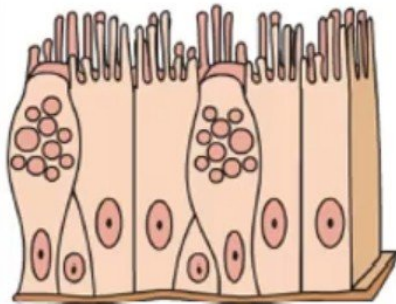
A Simple squamous



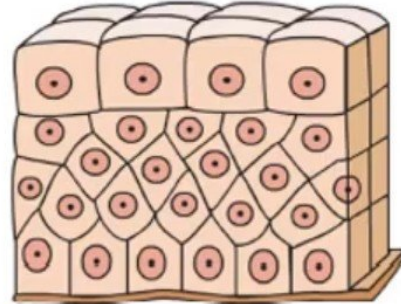
B Simple cuboidal



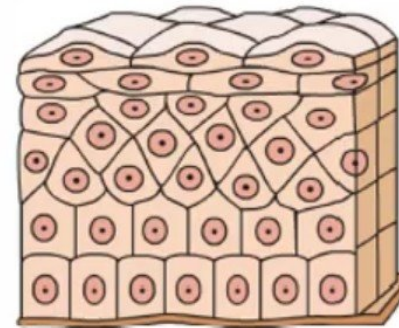
C Simple columnar



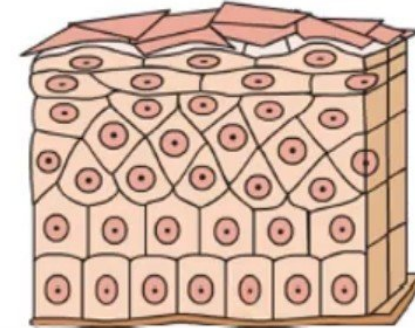
D Pseudostratified ciliated columnar with goblet cells



E Transitional

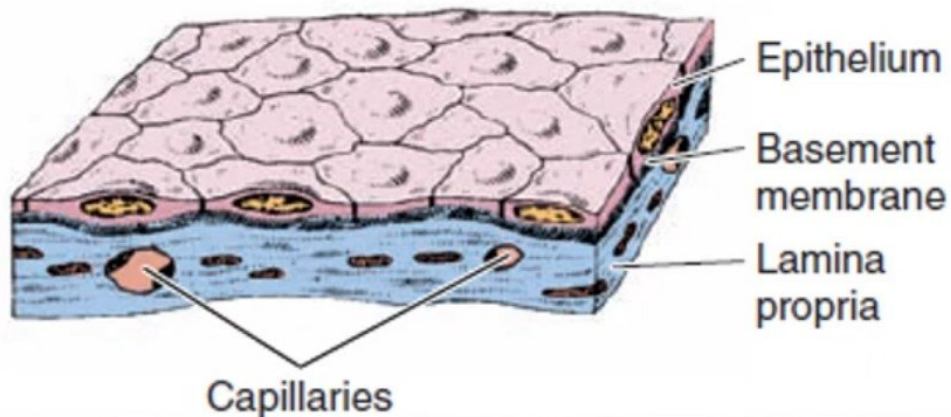


F Stratified squamous nonkeratinized



G Stratified squamous keratinized

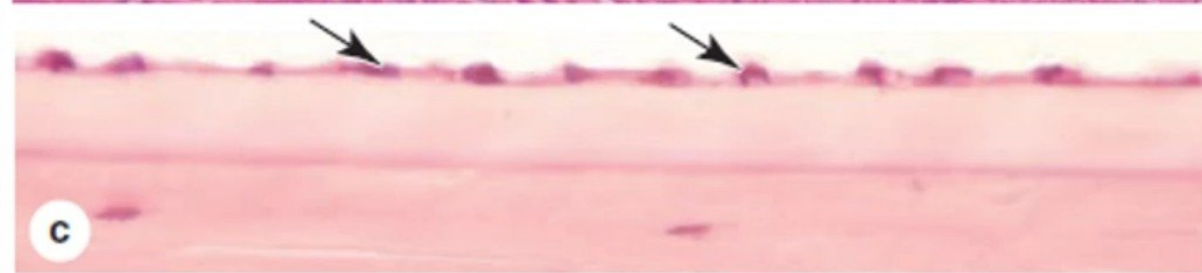
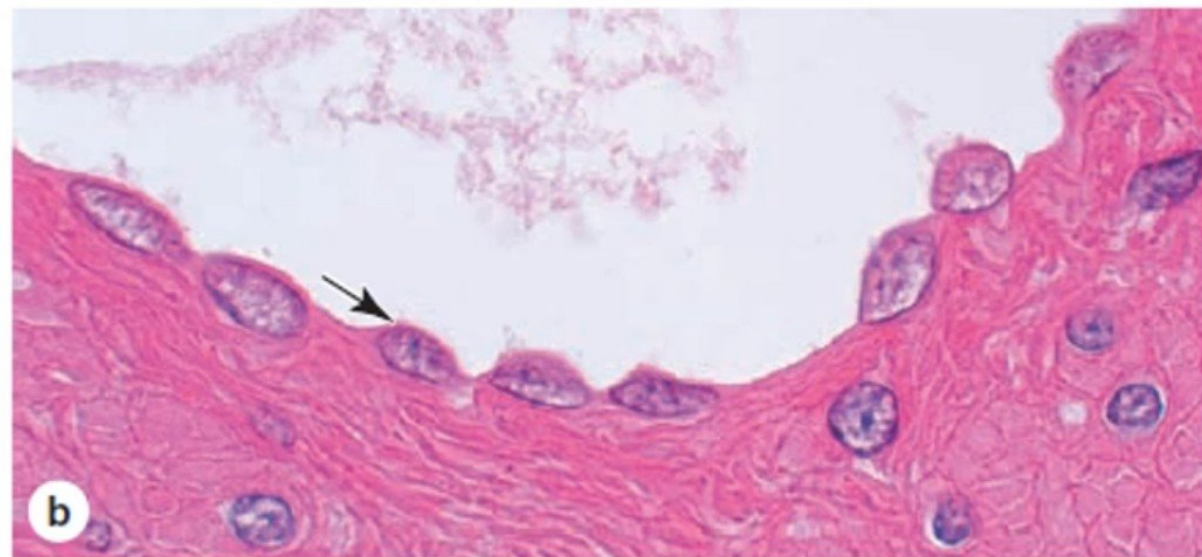
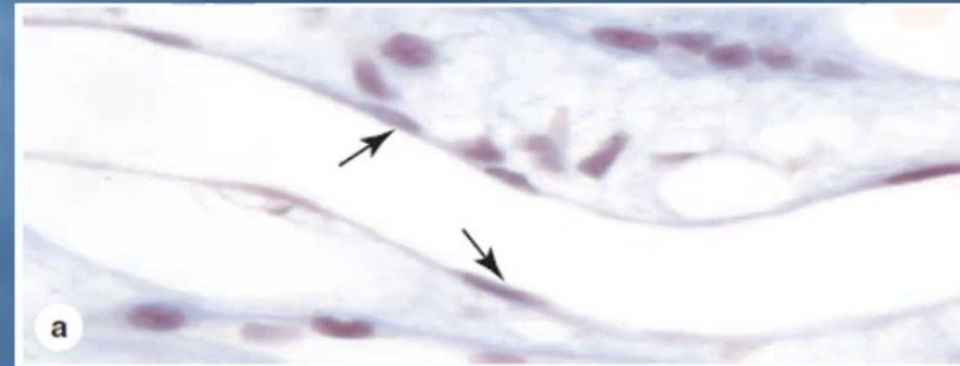
Simple squamous epithelium



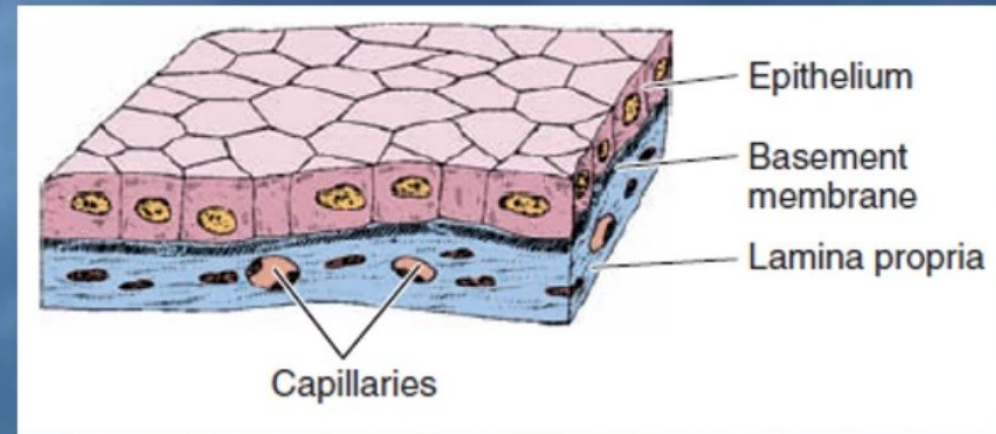
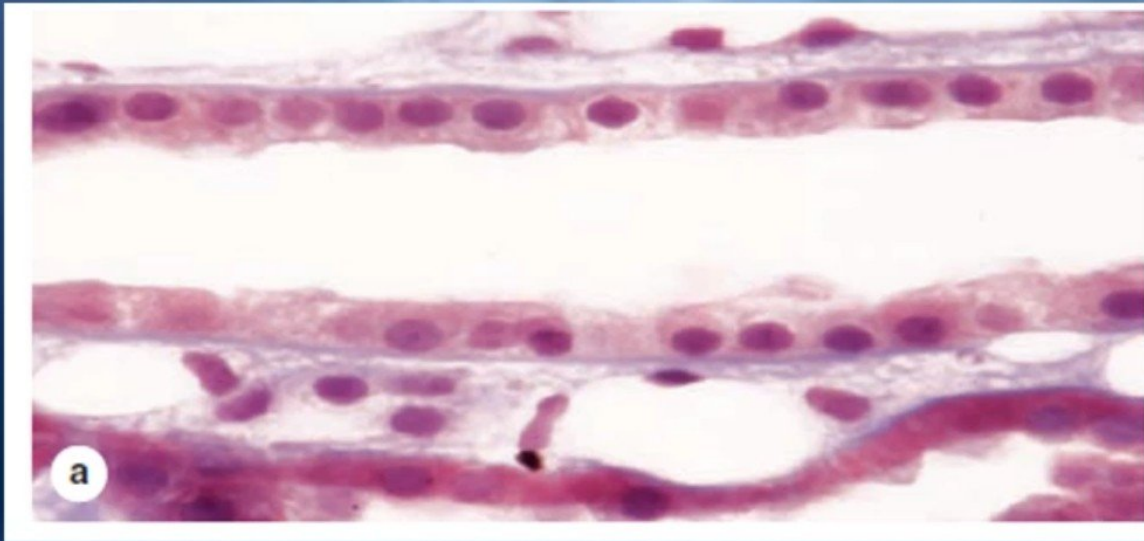
This is a **single layer of thin cells**, in which the cell nuclei (arrows) are the thickest and most visible structures.

Examples shown here are those:

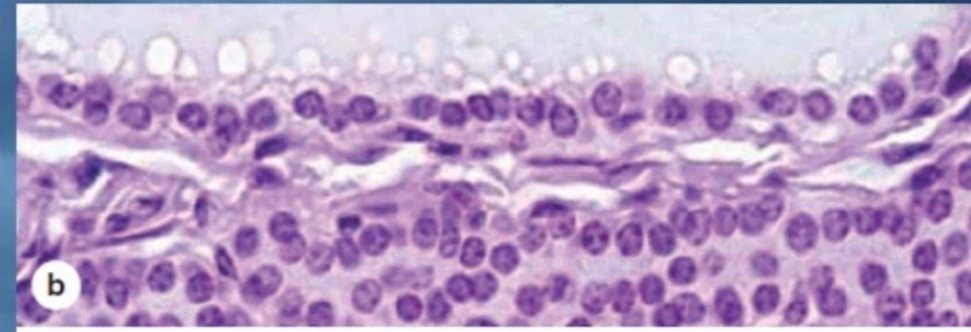
- lining the **thin renal loops of Henle (a)**
- covering **the outer wall of the intestine (b).**
- Lining the **inner surface of the cornea (c).**



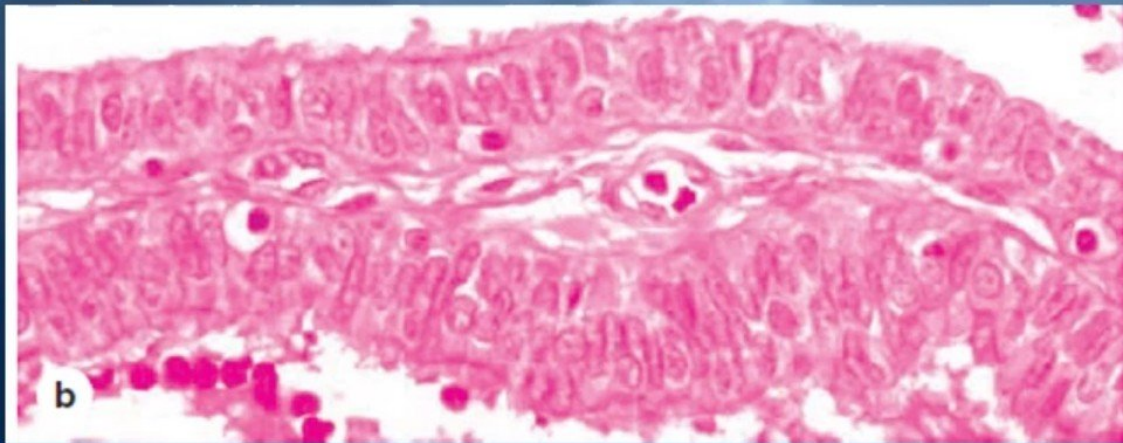
Simple cuboidal epithelium



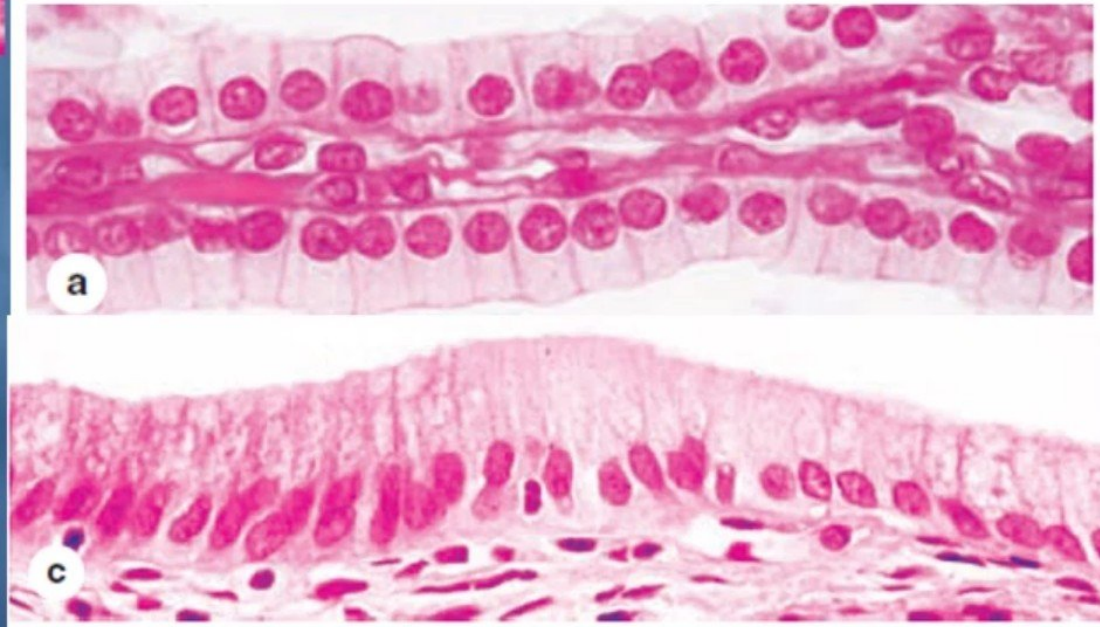
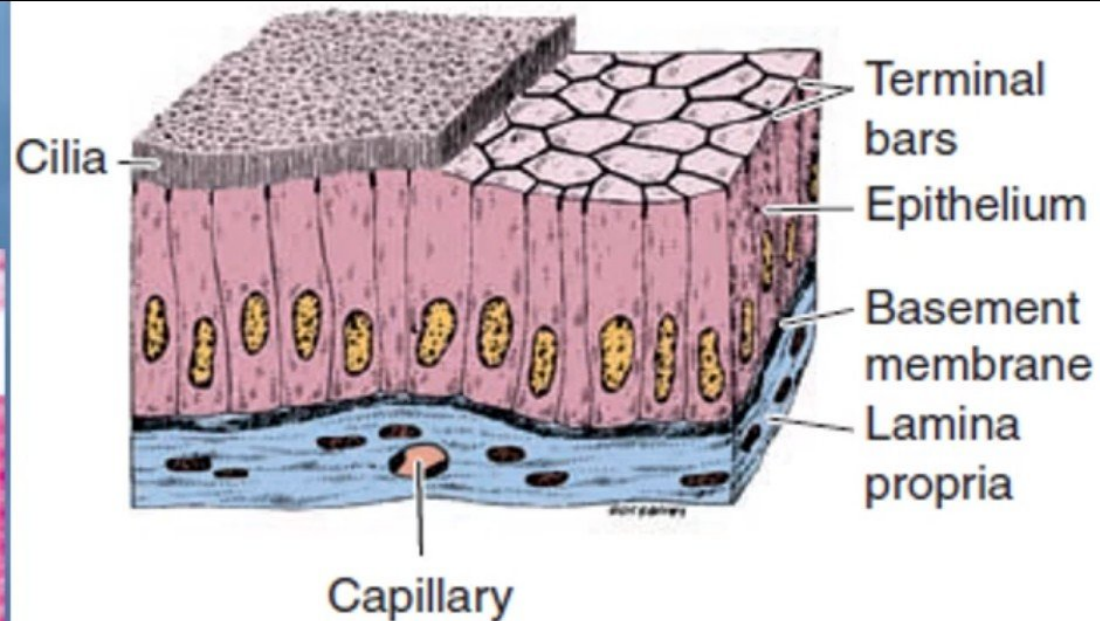
- Cells here are roughly **as tall as they are wide**.
- To be rich in **mitochondria** and other **organelles** for a **high level of active transport** across the epithelium.
- **Renal collecting tubule** (a).
- Large **thyroid follicle** (b).
- The thick mesothelium covering an **ovary** (c).



Simple columnar epithelium

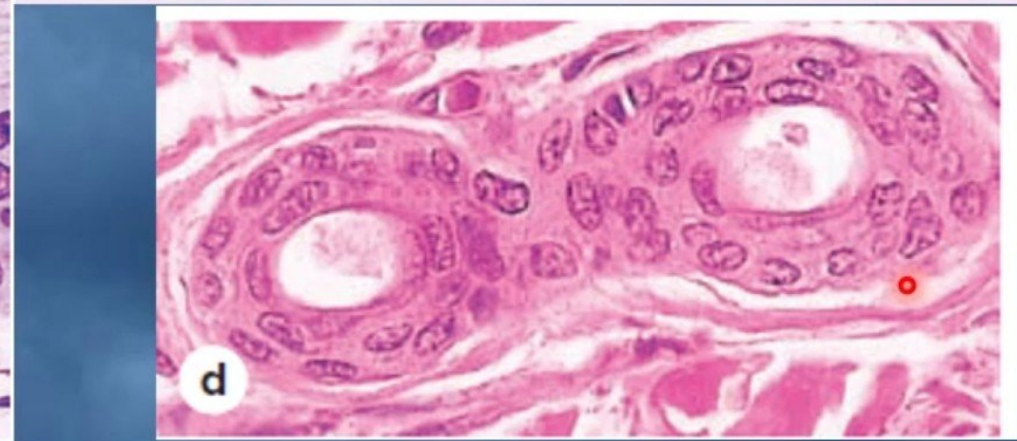
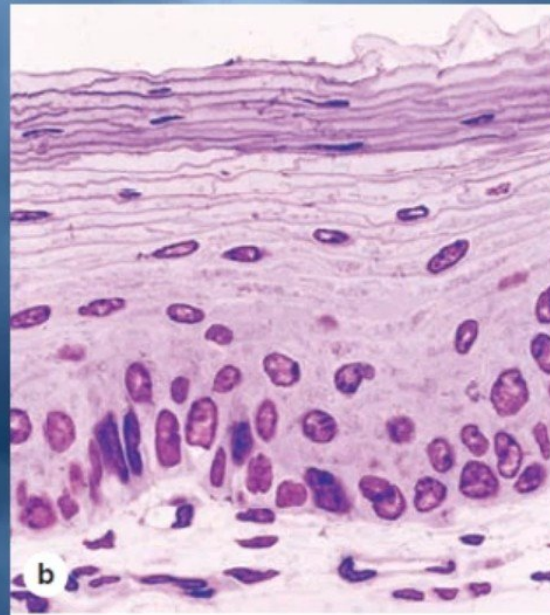
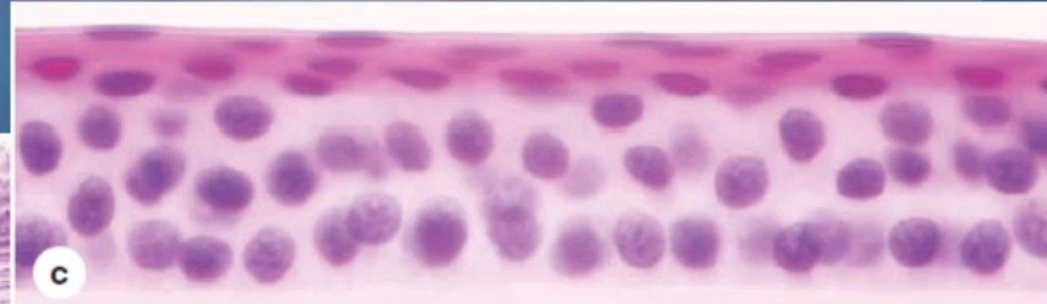
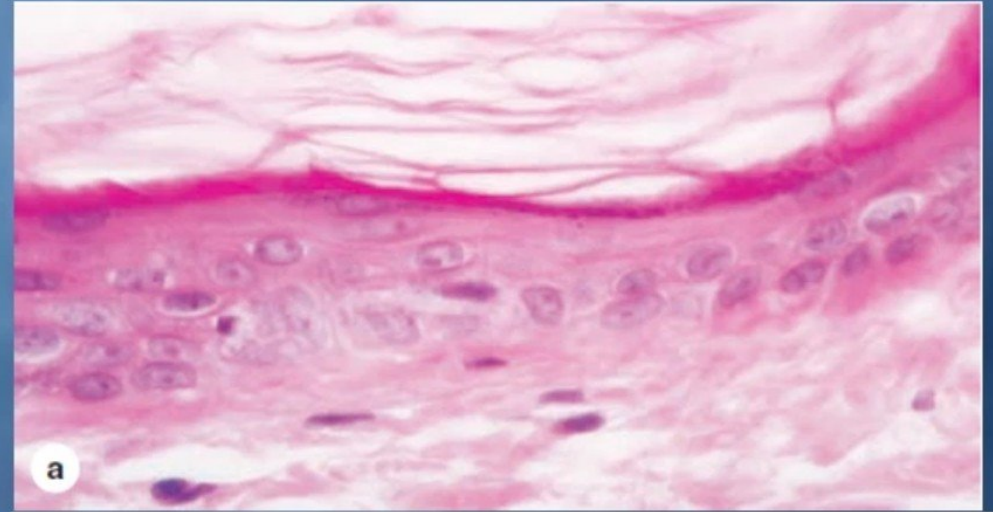


- Cells here are **always taller than they are wide**, with **apical cilia** or **microvilli**,
 - They are often **specialized for absorption**.
- The examples shown here are from:
- **The renal collecting duct (a).**
 - **The oviduct lining, with both secretory and ciliated cells (b).**
 - **The lining of the gall bladder (c).**
- All X400. H&E.



Stratified epithelium

- **Stratified squamous epithelia** usually have protective functions and water loss.
- These functions are particularly important in the **epidermis (a)** **Keratinized cells** are sloughed off and replaced by new cells from more basal layers, which are discussed fully with the skin.
- **Nonkeratinized epithelia** occur in many organs, such as:
 - **The esophageal lining (b).**
 - **outer covering of the cornea (c).**
- **Stratified cuboidal or columnar epithelia** occur in excretory ducts of certain glands, such as **sweat glands (d).**



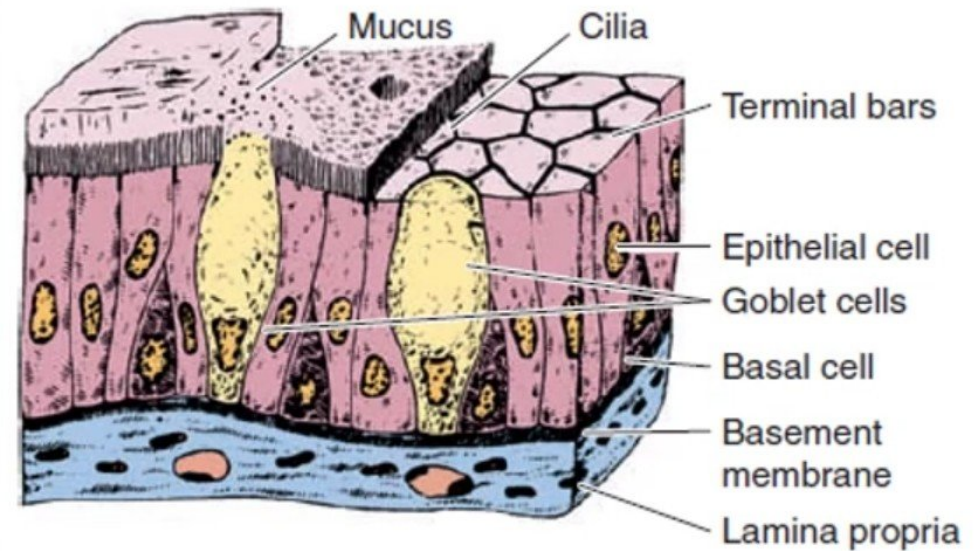
Clinical Note

- In **chronic bronchitis**, common among **habitual smokers**, the number of **goblet cells** in the lining of airways in the lungs often **increases** greatly.
- This leads to **excessive mucus production** in areas where there are **too few ciliated cells** for its **rapid removal** and contributes to **obstruction of the airways**.
- The **ciliated pseudostratified epithelium** lining the bronchi of smokers can also be **transformed into stratified squamous epithelium** by **metaplasia**.

Pseudostratified Epithelium

- Cells **appear to be layered** due to **irregularly placed nuclei** but **each cell touches the basement membrane**;
- May contain **cilia** – aids in movement of particles.
- **Function:** **protection, secretion, movement of mucus** and **sex cells**.
- **Location:** **respiratory tract, reproductive tract**.

The pseudostratified columnar epithelium of the **upper respiratory tract** shown here contains many **ciliated cells**, as well as other cells with their nuclei at different levels. X400. H&E.

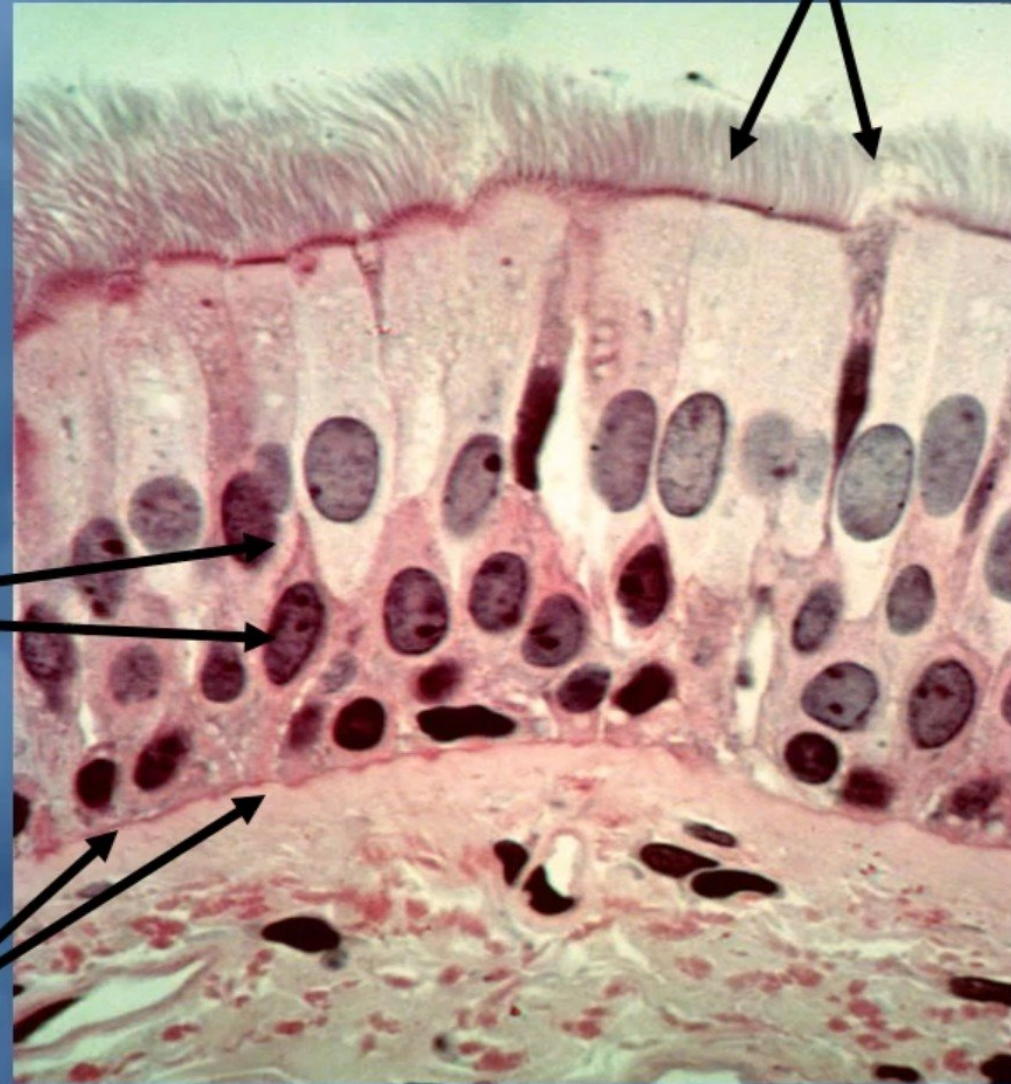


Pseudostratified Epithelium

Nucleus

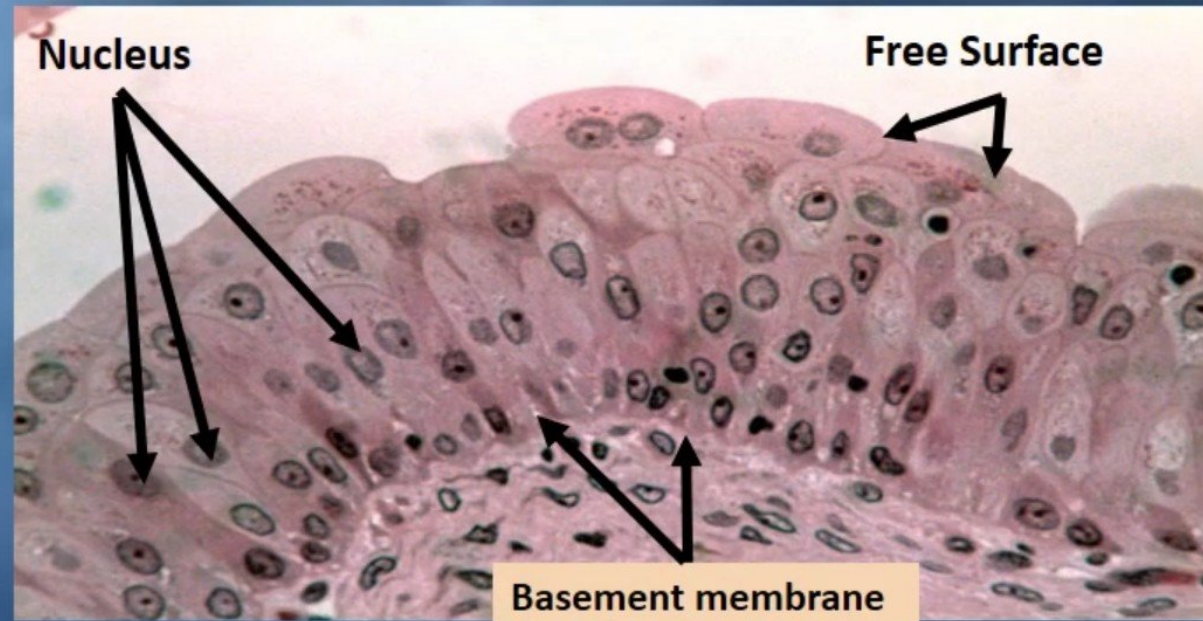
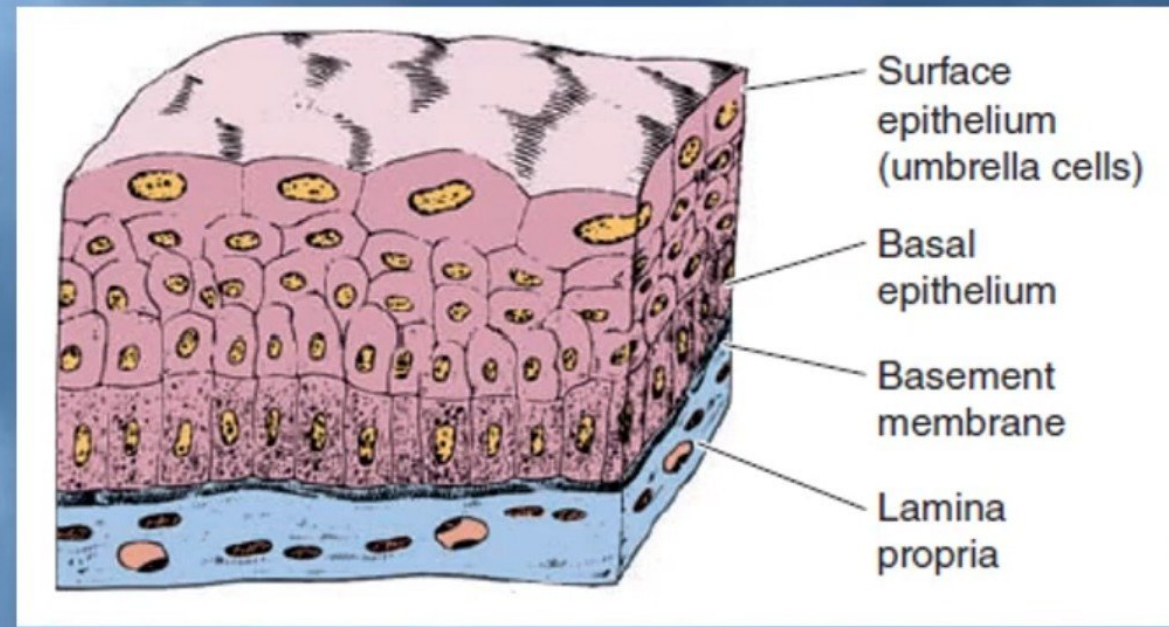
Cilia

Basement
membrane

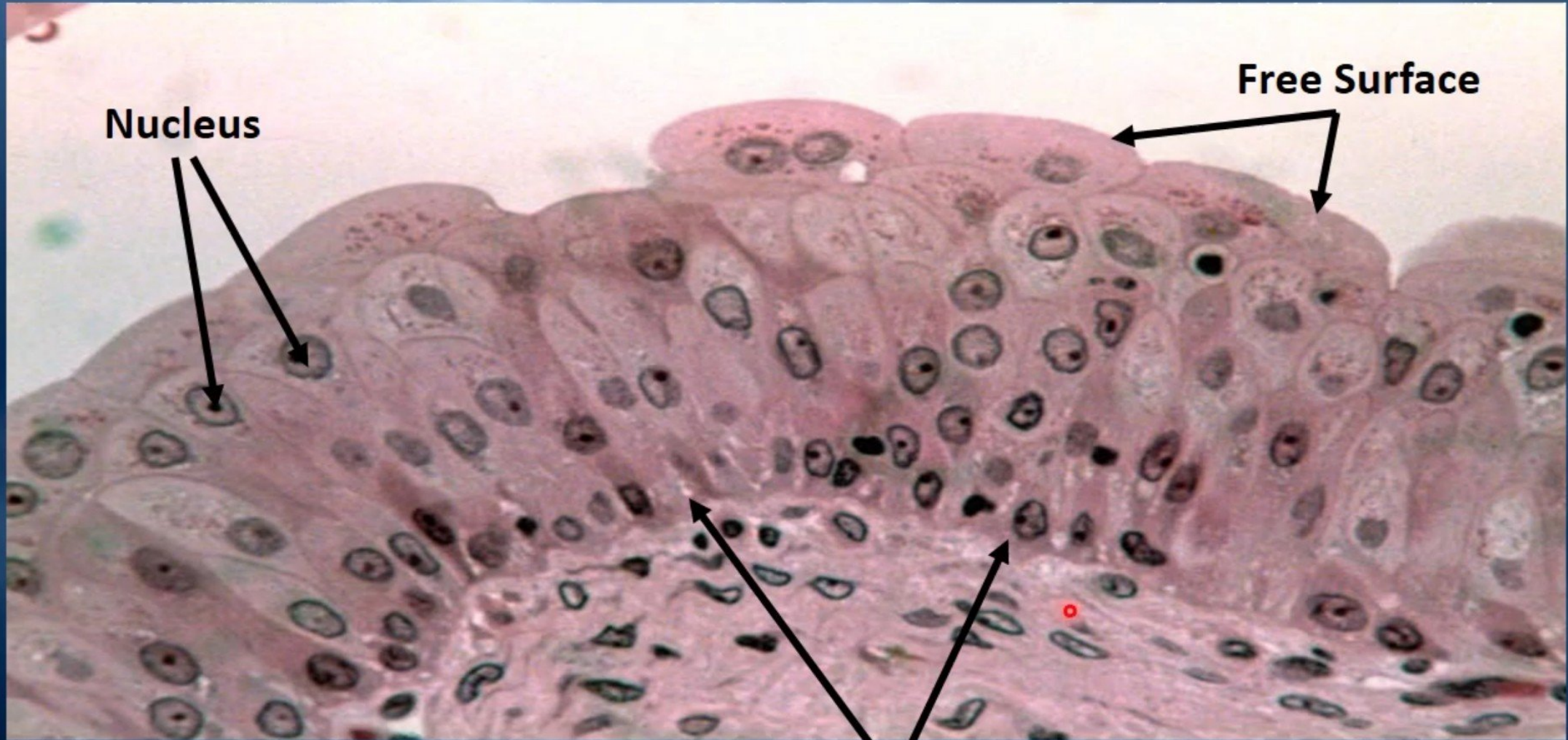


Transitional epithelium or urothelium

- The **superficial cells** are **rounded** or **dome-shaped**, and have **specialized membrane features** enabling them to **withstand the hypertonic effects of urine** and **protect underlying cells** from this **toxic solution**.
- Cells of this epithelium are also able to adjust their relationships with one another and **undergo a transition** in their appearance as the urinary bladder fills and the wall is **distended** and **stretched**.
- Location: **urinary bladder**, **ureters**, **urethra**.



Transitional Epithelium



Nucleus

Free Surface

Basement membrane

Common types of covering epithelia

Major Feature	Cell Form	Examples of Distribution	Main Function
Simple (one layer of cells)	Squamous	Lining of vessels (endothelium); Serous lining of cavities: pericardium, pleura, peritoneum (mesothelium)	1) Facilitates the movement of the viscera (mesothelium). 2) active transport by pinocytosis (mesothelium and endothelium). 3) secretion of biologically active molecules (mesothelium).
	Cuboidal	Covering the ovary, thyroid	Covering, secretion
	Columnar	Lining of intestine, gallbladder	Protection, lubrication, absorption, Secretion

Common types of covering epithelia

Major Feature	Cell Form	Examples of Distribution	Main Function
Pseudostratified (layers of cells with nuclei at different levels; not all cells reach surface but all adhere to basal lamina)		Lining of trachea, bronchi, nasal cavity	Protection, secretion; cilia-mediated transport of particles trapped in mucus out of the air passages

Common types of covering epithelia

Major Feature	Cell Form	Examples of Distribution	Main Function
Stratified (two or more layers of cells)	Squamous keratinized (dry)	Epidermis	Protection; prevents water loss
	Squamous nonkeratinized (moist)	Mouth, esophagus, larynx, vagina, anal canal	Protection, secretion; prevents water loss
	Cuboidal	Sweat glands, developing ovarian follicles	Protection, secretion
	Transitional	Bladder, ureters, renal calyces	Protection, secretion
	Columnar	Conjunctiva	Protection, secretion

Thanks